

This is not AI generated , this doc contains my understanding of AI .

AI is something which can be used to accelerate any of desk work - in future its gonna be used in 3d works as well **boston dynamics** is making good progress and china as well , and not only accelerate the core idea of AI is basically a neural network , with multiple layers as the same way as our **brain**..

The only thing claude code lacks compared to our brain is that **long term memory** and it tries to **hallucinate** /guess the answer most times unlike us . And during my time at **Rubrik** i have resolved the bridge and made use of agentic AI most with long term memory called **knowledge graph** and also having the agents work in a particular way - basically **mini agentic framework** .

Or here is an ex:-

Lets say someone is working on codebase , and a new task needs to be done , where a similar task has already landed on master - in this case AI can do a better job at replicating things - since its good at finding similarities and wt not writing code , in this case all we need to do prompting .

But if the task is going to be new and there were no similarities with the codebase , in this if u let AI do the tasks most probably it tries to guess or we can put it this , tries to get the **best answer** rather than **accurate** answer . and here is the actuall bottleneck , all the AI companies are not only trying to minimize this by training the models with the data they have scrapped from free tier users and fetching lot of webpages from **clean internet** . In this case we have to design the things in order to finish the task , and let AI understand wt exactly we are trying to do , basically do POC and then let the agents do their work .

AI is mostly of 3 things , **Compute ,Algorithms** and **Data for Training**

Compute is mostly handled by NVIDIA / AMD - they have done a great job at building chips which could help models run locally . Oracle and Elon companies buys the GPUs from nvidia basically and recently Anthropic has recently made a deal with Elon **1.5BPM** worth of compute on colosso II engine .

Algorithms are mostly handled by the companies like Anthropic, OpenAI, Deepmind from US and from china they have build great model which are in par with fable 5 like **kimi k3 (Moonshot)**, **GLM 5.2** and if u have seen japanese model like - Sakana Fugu , instead of building a new model they built a mini model which basically tells in the next transcript which model in the pool will pick this iteration and wt does this model do , whtr its **research / impl / verifier** .

And the rumour is that Anthropic has found its way to **RSI** (Reccursive Self Improvement) But someone else has also tested this - Basically letting AI decide its future [AIDE2](#). If u have watched **Point Of interest** webseries U might understand where exactly are we going on with AI in form of LLMs in the world , and wt exactly happens when it becomes autonomous .

Now coming to **AI warfare** between US and china , these are the two main competitive countries and of course india is a silent killer .

Basically most of the recent chinese models are open source , by making the models open source which are on par with fable or the most closed models , china is degrading the AI revenue/economy of US companies . If u a have seen recently **Databricks** has started self hosting kimi K3 models .

Now coming to Attacks by AI - **Hackers** can **jailbreak** the models by bypassing the safeguards this is a bit tricky compared to **prompt injection** - famous injections from 3 years ago would be **DAN(Do anything mode)** , and role based . Basically letting AI write the Payloads is the one last thing any use case of it , AI is built to be helpful and accelerate the good work , by bypassing the guards the core principle is getting deflected.

Now coming to mythos , i dont know how they have built such a powerful model , my guess would **fine tune** the model with the **vulnerabilities** , this way the weights will get designed to be good at finding vulnerabilities , and in this case mythos not only finds, it exploits teh vulnerabilities as well , and one recent incident would be when **OpenAI** is testing their agents **off the grid** (not on network) , the agents found their way to network and hacked **Huggingface** i think .

The way that we need to fine tune a model would be like this i would say :-

Lets say a trader thinks this way for example he mostly depend on support and resistance lets say , and he some how predicts with his instincts that now its going to moved out of **Support and resistance** and its going to be UP/DOWN based on parameters like momentum or wt not , now we need this way of **thinking** as training data for teh model to fine time the model , so that it could think better , rather than just telling it predict and just fine tune the data . we have to tell when the situation is liek this this is wt i would do .

Now we move onto agents - agents are like ur mini workers , like u tell them how to do the work exactly or give them ur direction or ideology and they will basically do the work for u .

This is wt i have **build** during my time at **rubrik** :-

We have something like multi agents - where u have like a **pool** of agents waiting for them to be called from amin session , and each agent is assigned a role one for **research** and one for **checkpointer** , **Implementor**, **Tester** , **logger** and finally one main session - **HEXAGON**

- 1 . **research** - is designed to understand the codebase
2. **Implementor** - is designed to break the big task and get the code preview on disk and then on my approval via main session it will implement
3. **Tester** - for testing
4. **Logger** - looks at all agnets and logs about each agent which tool it called
5. **Main session** - this is because , when we try to give some context to any session , we tend to give minimal context of laziness , like telling it briefly , and in this case the main session can enrich the context that u sent to pass onto corresponding agent from **POCs**

And for long term memory u can do this basically log every session that u have done into md files in obsidian or any ide , and u can tell AI to attach the previous session or POC's md that are relevant to this md

I have few other ideas like **cross session intelligence** , and also making a proper **handoff** btw sessions . and some **autonomous** agents with **good distinguishing** , acting as **firewalls** in case of any live **breaches** of security .

And building agents are **purposefully** designed to deal with **git software** - like resolving commits or **raising PRs** and wt not , and **agents** who are **designed** to code , and some to do **research** in codebase.

And every software so far we build is designed in a way to be **human** friendly , like search **algos** or UI's , like take **glean** for example , ppl use glean to access their companies **internal knowledge** , but since we are going to **ai world** , we should build **software** which is fine tuned to be **agentic/AI** based . like having an **agentic memory** which is used by agents to **access** relevant info . More of like an **agentic language** , like any normal lang **english** or **telugu**.

And also **logging** every session every **employee** had , so that instead of models do **thinking** and finds the **contents** via search , they can find them in **KG's** before even start thinking , basically tell **models** to look at our **customized cache** rather than models taking a **random** guess and **making** them grep **across** codebase(which is huge cost) and having a customized **harness** which is built **efficiently** for ur usecase.

And having genetic work trees , and **having a pool of agents** which basically **validates** other agents , and **routing agents** , basically an **agentic** network , you **could** say. And layers of agents , which would validate, like local cache , CDN and server cache and then server equivalent.

And can we do trading with LLMs , there comes a bottleneck pf latency which is teh main goal that HfTs will try to optimize .they have to fill the model with enough parameters so that it wont suggest wrong info and also doesnt payoff latency.

Currently we can see **significance** web traffic due to **agents fetching** data , in future is it going to be like **agentic network** with its **own** protocols - a network space where agents could **communicate** , would there be any loop in for **humans** (if so it should be via agents only wt if those agents went rouge) , how can we figure out agents didnt went **rouge** .

So far this is just a **preview** of AI as how it can be useful technically for **non AI companies**.

Now with AI there is a large scale disturbance now wt can be done with **AGI/ASI** - Now imagine wt would happen with **Quantum neurons** . So far the models use normal neurons , wt if quantum neurons comes into role.

Now the big question is AI better than the human brain ?

My answer for this question would - its not that AI is better than Human , **Human brain** is also a **neural network** ,and each person has their own way of **quick fine tuning** their brain

with knowledge - and one thing AI has it can easily **grep things** in codebase , but if some person who had worked on a codebase for 10 yrs human would be quicker than AI . Its all about the weights in our brain , and how **good neurons** gets **enhanced** and bad neurons get destroyed during sleep.

If u see this example , in india if u find a **snake** u would run like **hell** , until the distance is so far . now imagine how do people in **china** catches snakes with bare hands without any fear even venomous , its because the weights in our brain are developed based on our environment , we were told during our early stages to act this way or interact this way when u greet some one ,

Actually **bioengineers** did a experiments , they put some under 2 yr kids in a big cube and they left non venomous snakes in there, and teh kids tried touching , from this u should be able to understand its not the AI is better than human brain , we are not using teh brain in the right way .

And model instance would take **10 times the power** our brain takes and it still needs the **water to cool it self** unlike our bain can kept around room temp, and the **compute extent is 1/10 (no of ops)** of wt happens in our brain , the only difference is that fine tuning model is easy but thats not the case with brain .

And Not only this , Fable 5 was able to help solve a myth problem u might know by [jacobian Conjecture](#) . In future how well are the models gonna do .

And recently **Anthropic** tied up with Ode led by **Blackstone**, its basically the actual usage of Ai and the current usage of AI at the Saas or trading companies is way big , now ode with anthropic is basically they send their engineers to customer companies and help them build ai tools that are useful for company.

And to efficiently do the computation ppl have come up with smthng called **sparsity** , basically its just that only the related branch of neural network get activated , not the entire network , so this no:of ops gets decreed and also the load on cores would be less.

One good question might be how can we minimize the gap btw human brain and models , in terms like memory or wt not ?

Currently the models are good at helping humans figuring out , wt would advanced models look like , would they be able to autonomously do work by themselves, without any human guiding ? If so, how would it help for ppl who are employing at non AI companies ?

And also AI is build to act as a god guide , but people recently have been misusing a lot , in a way that **anthropic** has found lot of bad feelings in their claude models brain , and also china is conducting **MMA** fights btw **robots** .

RSI ref - <https://www.weco.ai/blog/first-evidence-of-recursive-self-improvement>

The most complex thing in world is our **brain** (atleast in the things ik) , if ppl have mastered on how to use it , ull find 100 **Elon musks** in world.

And in any use case of AI , if it did any mistake or oversaw any case , we should resolve that , not only resolve that , we have log these king of this in the agent memory , so that in future it wont repeat , and also we need to ask agent why did it this way like oversaw any usecase , and instead of just fixing , we have to let it realise , wt can be done to improve .

Question to bioscientists

Is it possible to **extract** the **weights** from the brain - if so , someone could **monopolize** a **company** with this idea , **basically** having a global **knowledge** , and provide services . to stop this from **happening** wt can be done .

And can we **artificially** accelerate the **brain** enhancing - like enhance the **good neurons** which help them . If this can be done , wt can we do to bad neurons wt is the **downside** of not having **bad neurons** or less than the **threshold**. And how can we stop ppl from enhancing **bad neurons**?